SELF DRIVING CAR NOTEBOOK

Author: TJ

UDACITY Nanodegree

1 Introduction

2 Machine Learning
- Machine Learning Workflow

3 Extras
- Modeling

SECTION

Introduction

SECTION

Machine Learning

SUBSECTION

Machine Learning Workflow

ML METRICS 1/ -

Metric	Definition
True Positive (TP)	Contains item and predicts item
True Negative (TN)	Does not contain item and predicts no item
False Positive (FP)	Does not contain item and predicts item
False Negative (FN)	Contains item and predicts item

ML METRICS 2/ - CLASSIFICATION

Metric	Equation	Definition
Precision	$\frac{TP}{TP+FP}$	Error rate of algorithm when it's predicting an iter
Recall	$\frac{TP}{TP+FN}$	Error rate when the picture is correct
Accuracy	$\frac{TP+TN}{TP+FN+FP+TN}$	Total error rate

ML METRICS 3/ - INTERSECTION OVER UNION (IOU)

- Create bounding boxes for prediction and observation (or truth)
- Ratio of intersection of bounding boxes to union of bounding boxes
- 50% ratio considered threshold for true positive

DATA ACQUISITION AND VISUALIZATION

- Most time spent at this stage
- [TERM] Domain Gap - an alg trained on one specific dataset may not perform well on another

EXPLORATORY DATA ANALYSIS (EDA)

- Should spend a significant amount of time visually examining dataset
- [TERM] Domain Shift - Conditions that affect ML alg ability to analyze
  - Examples: weather, light, sensor change/processing, environment
- EDA can help tune data analysis for domain shift
- Vision problem can require looking at 1000s of images

CROSS VALIDATION

- Ensures alg will perform well in any environment
- Evaluates the generalization ability of the model
- Concepts
  - Overfitting
    - Model does not generalize
    - Differentiate training vs test datasets
  - Bias-variance tradeoff
    - Hard to create a balanced model
    - $TestError = Var + Bias + \epsilon$
    - Variance - sensitivity to training data, bias - quality of fit, test error - error rate on test datasit
    - Reducing variance will increase bias
  - Cross Validation
    - Evaluates how well a model generalizes
    - Be careful of data leakage

SECTION

Extras

SUBSECTION

Modeling

VEHICLE MODELING

- Kinetic model
  - Simplification of dynamic model
  - More tractible
- Dynamic model
  - Aim to encompass dynamics of model
  - More robust

STATE

- Position, orientation, velocity
- In this example
  - [x,y,psi,v]

BUILDING A KINEMATIC MODEL

- Actuators - Control inputs
  - eg steering wheel, pedals
  - in example, these are delta for turning and a for acceleration

FOLLOWING A TRAJECTORY

- With self-driving car
  - Perception - localization - path planning - control
  - Usually use third degree polynomial for planning
  - Localization compares model to map with SDC
  - Path planning uses map, veh loc, environmental model

ERROR 1/

- Must measured and expected values
- State becomes [x,y,psi,v,cte,epsi]
- CTE - Crosstrack error
  - Low complexity heuristic
  - Difference between intended route and current track
  - For next cte

$$cte_{t+1} = f(x_t) - y_t + (v_t * sin(e\psi_t) * dt)$$

  - Where $cte_{t+1} = f(x_t) - y_t$ is curr cte and rest is error due to veh mvmt

ERROR 2/

- Orientation error
  - Difference between desired and curr orientation
  - $$e\psi_{t+1} = e\psi_t + \frac{v_t}{L_f} * \delta_t * dt$$
  - Becomes
  - $$e\psi_{t+1} = e\psi_t - \psi des_t + (\frac{v_t}{L_f} * \delta_t * dt)$$
  - Where $e\psi_{t+1} = e\psi_t - \psi des_t$ is the curr orient error and rest is new error caused by movement